

Process Control

Final Lab Report #5
CHE 381 - Unit Operations Lab I
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Group #3

Javier Eduardo Mendieta
Omar Zahdan
Omar Takroui
Joel Finlon

1. Executive Summary

The purpose of this experiment is to control the water level in a vessel by varying the pump speed both manually and automatically. A drain valve is opened and closed to provide disturbance to the system in order for corrections to take place, namely the pump changing its speed. The system produced proportional (P) and integral (I) values for maintaining the water level at a set point [1].

Manually varying the pump speed was found to be less efficient than using the P and I automated methods. Stability was returned to the system quicker using the P and I methods than the manual method, as predicted. The higher the proportional band the slower the system response time. The method using both P+I produced significantly higher proportional band values than the other two methods [1].

2. Introduction

The objective of this lab is to control the water level in a process vessel by varying the flowrate of water coming into the vessel by changing the pump speed. In Experiment A we change the pump speed manually, while in Experiment C the pump speed is controlled using a proportional controller (P) and a proportional + integral (P+I) controller. By doing this, we can observe the effects of manually controlling the pump speed compared to using a P or P+I controller [1]. These principles are fundamental to the subjects of process control engineering and control systems engineering. By performing these experiments, we not only learn the functionality of control systems on a process unit, but we also observe the effects of different types of controllers in a level controlling process [2].

We used a PCT50 level controlling process with water being stored in a lower sump tank which is then pumped up to a process vessel using a variable speed centrifugal pump. The water level inside the process vessel is measured by a pressure sensor attached to the floor of the vessel. Also in the floor of the process vessel are two outlets in which the water can flow out of the vessel and back into the sump tank [2]. These are the main outlet and the solenoid valve which are both operated by hand operated ball valves to control the flow through them and fitted with orifices. The setup is hooked up to a PC with the software necessary to monitor and operate the process. The process can also be operated using a PCT54 PID Controller or a PCT55 Programmable Logic Controller connected to the electrical interface [3].

Level control is used very often in industrial applications. Processes that use hoppers, tanks, or other types of holding vessels typically will use level control on that vessel to ensure overflowing or dry running does not occur. It is also very useful to know the fill level of each tank in a process, as you may not want the material in the vessel to be in there for a long period of time to avoid corrosion. Or, you may want to ensure a vessel is completely empty prior to

initiating cleaning [4]. Whatever the case may be, level control is vital to many industrial processes and being familiar with the control system behind this process is very useful.

3. Theory

The PCT50 functions with respect to the fundamental principles of process control engineering. The system measures level as the manipulated variable of a single PID loop with Proportional Band, Integral Time and Derivative Time as correcting terms [5].

Represented in the PID are the following variables:

- Proportional Gain (K_p)
- Integral Gain (K_i)
- Derivative Gain (K_d)

Consider the error (E) as the difference between the set point (SP) and the process variable (PV):

$$E = SP - PV$$

- Instantaneous Time (t)
- Integration 0 to t (τ)

$$u(t) = K_p e(t) + K_i \int_0^t e(t) dt + K_d \frac{de(t)}{dt} \quad (\text{Equation 1})$$

Proportional Band

Proportional Band term represented as a percentage proportional to the current level. The response is adjusted by the product of error (E) and Proportional Gain (K_p).

$$K_p = \frac{100\%}{\text{Proportional Band \%}} \quad (\text{Equation 2})$$

The sensitivity of the PID controller is manipulated through the proportional gain (K_p). As seen in the above equation error (E) plays a significant role for the value of K_p . A large K_p resulting from large error value will result in an overly sensitive system while a low K_p yields decreased responsiveness[5]. A process controlled by proportional gain will generate an error requiring integration for the final steady state.

Integral Term (I)

The Integral term represented by the equation below corrects for an accumulation offset proportional to the magnitude and duration of a given error.

$$I = K_i \int_0^t e(\tau) d\tau \quad (\text{Equation 3})$$

The integral term is set forth to eliminate errors present in a proportion only controller by directing level movement toward the set point (SP). The product of accumulated error and integral gain is added to the controller output [1].

Derivative term (D)

The derivative term is a result of the changing error over the process time. The calculated rate is then multiplied by the derivative gain. Derivative Gain (K_d) serves as a value for the overall derivative term effect on the process level.

Actions of a derivative term on a process variable may improve the consistency of the system by acting in a predicting manner to restrain the system within given setpoints. However, industry does not typically implement the use of derivative action due to the possible “noise” created by high frequency inputs. An implemented filtering term accompanying the derivative term may stabilize fluctuations to better model a level system [1].

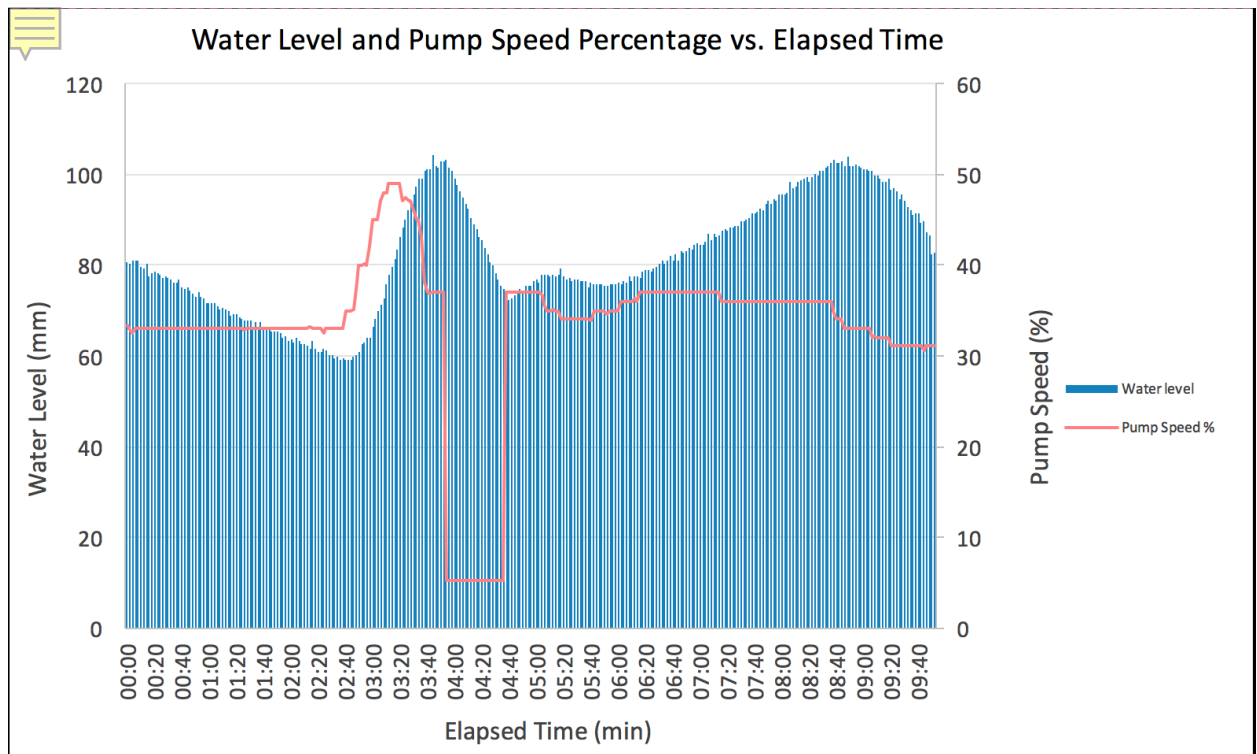
4. **Results**

Experiment A:

The first experiment of this laboratory required the water level of the tank to be manually regulated by manipulating the flow rate of water that enters and leaves the tank. The main objective of regulating the flow rate is to reach steady state (at 80 mm), from which it would be possible to determine how any disturbances in the system would affect the water level.

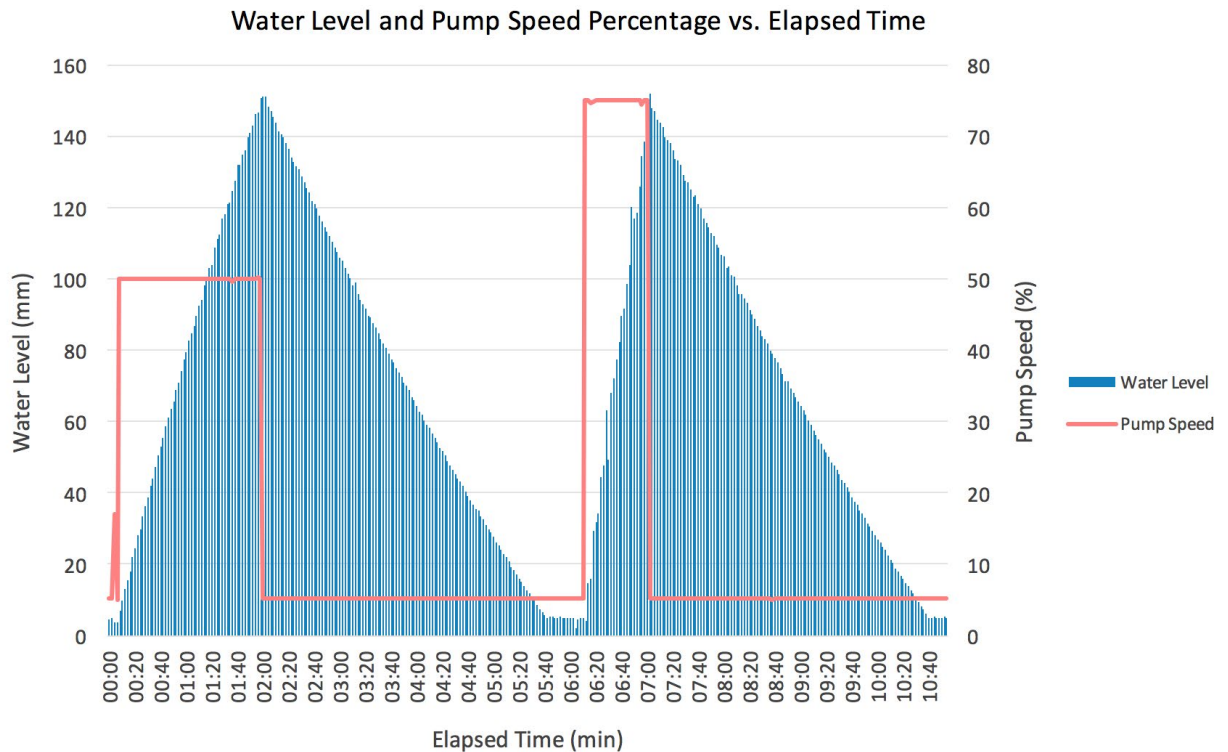
To achieve this, the CV1 valve had to be manually manipulated to balance the flow of water while the speed of the pump was set to approximately 33%. Then, the required disturbance (solenoid valve opened to 80 mm) was applied to the system decreasing the level of water since most of it flowed out. Next, the solenoid valve was set back to its original position while the speed of the pump was increased to 50 %. From here, the level of water increases to 100 mm. Then, the pump speed % is decreased to 10% and the flow of water decreases to 75 mm. Finally,

when the pump speed is regulated back to 33%, the water flow rises back to 100 mm and then it reaches steady state at 80 mm; which was the initial desired level.



Graph 1 - Water Level as a Function of Time Elapsed for Manually Controlled Pump at Initial Fill of 80 mm.

As seen in graph 1, the water tank was initially filled at 80 mm (solenoid valve closed) at time equal zero and pump speed of 33%. As soon as the valve is opened, the water level decreases to approximately 60 mm while keeping the pump speed at 33%. When the pump speed is increased to 50% and the valve is closed, the water level rises up to 100 mm as time goes by.



Graph 2 - Water Level as a Function of Time Elapsed for Manually Controlled Pump at Initial Fill of 0 mm.

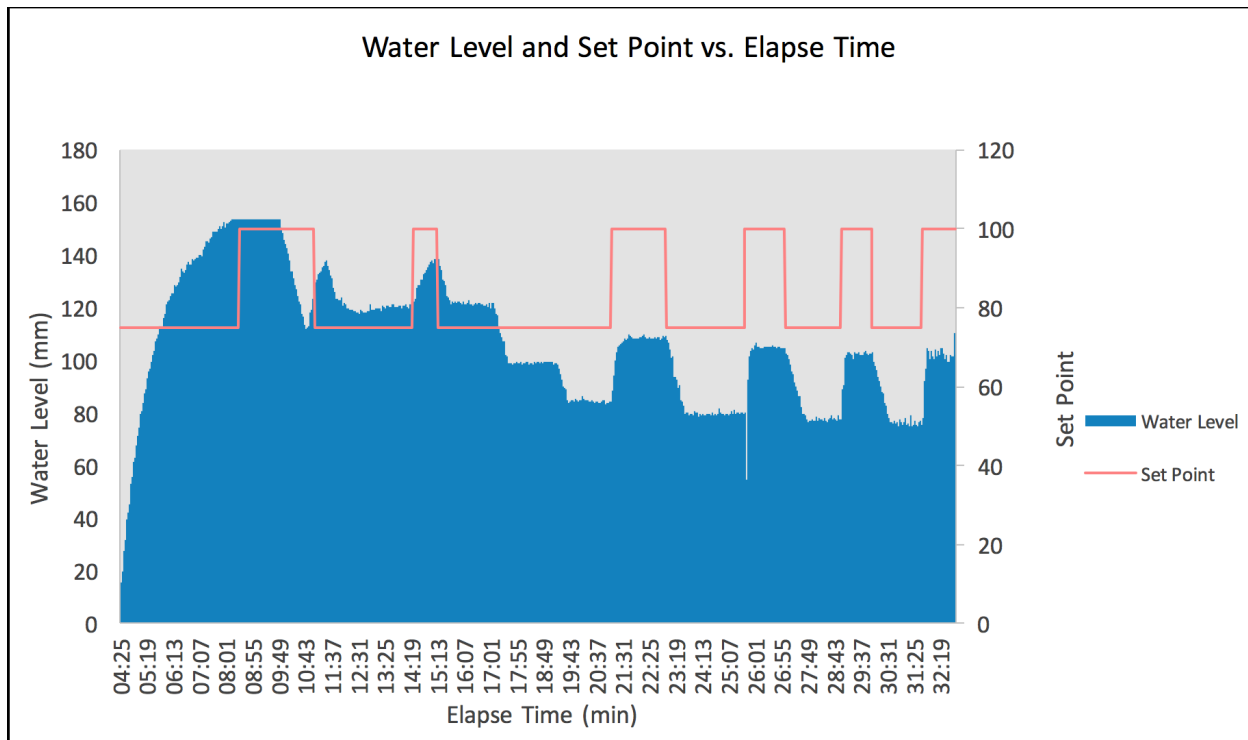
For Graph 2, it is possible to notice how the pump speed and the water level are correlated as a function of time. This is initially seen at time equals 0, while the water level is also equal to 0, the pump has not been run yet and the solenoid valve has not been opened. As soon as the pump is set up, the water level increases at a rate of approximately 20 mm per 15 seconds. The tank is filled all the way up to 150 mm before the water level starts to decrease as the solenoid valve is opened and the pump speed is reduced to 5%. A similar trend is displayed for the second half of the experiment, in which the pump speed is set up to 75% for one minute while the water level rises to 150 mm and the valve is closed. Finally, the valve is opened once more, from which the graph displays a decrease in level at a rate of 20 mm per 15 seconds all the way down to 0 mm in a lapse of 4 minutes and at a pump speed % of 5.

Experiment C:

The second experiment of this laboratory consisted of an automatic operation of the system. This was carried out by automatic manipulation of the Proportional and Integral values (P and I respectively). The main goal of this second portion is to analyze how the system would respond to any possible disturbances that may be present while the tank operates. These disturbances are generated by changing the set point, the pump speed percentage, and how much the solenoid valve is opened. Finally, these disturbances are recorded and conveyed in the

following graphs for two different sets of configurations. The first configuration is only for P, while the second configuration requires P + I.

For the P configuration, the proportional percentages were tested at 2%, 5%, 10%, 20%, 50%, 100%, and 200%. For each of these setups, the set point was established to be at 75 mm, initially. As time went by, the set point rose up to 100 mm for a water level of 150 mm. Then, the water level decreased to 120 mm when the set point was 75 mm. This can be seen in the graph 3, where the level of water oscillates as the solenoid valve is continuously opened and closed automatically by the built-in software. Along with this, the pump speed is automatically controlled whenever the valve opens and closes; therefore, generating measurable disturbances on the water in the vessel.



Graph 3 - Water Level as a Function of Time Elapsed for Automatically Controlled Pump at Initial Fill of 0 mm and Initial Set Point of 75 mm.

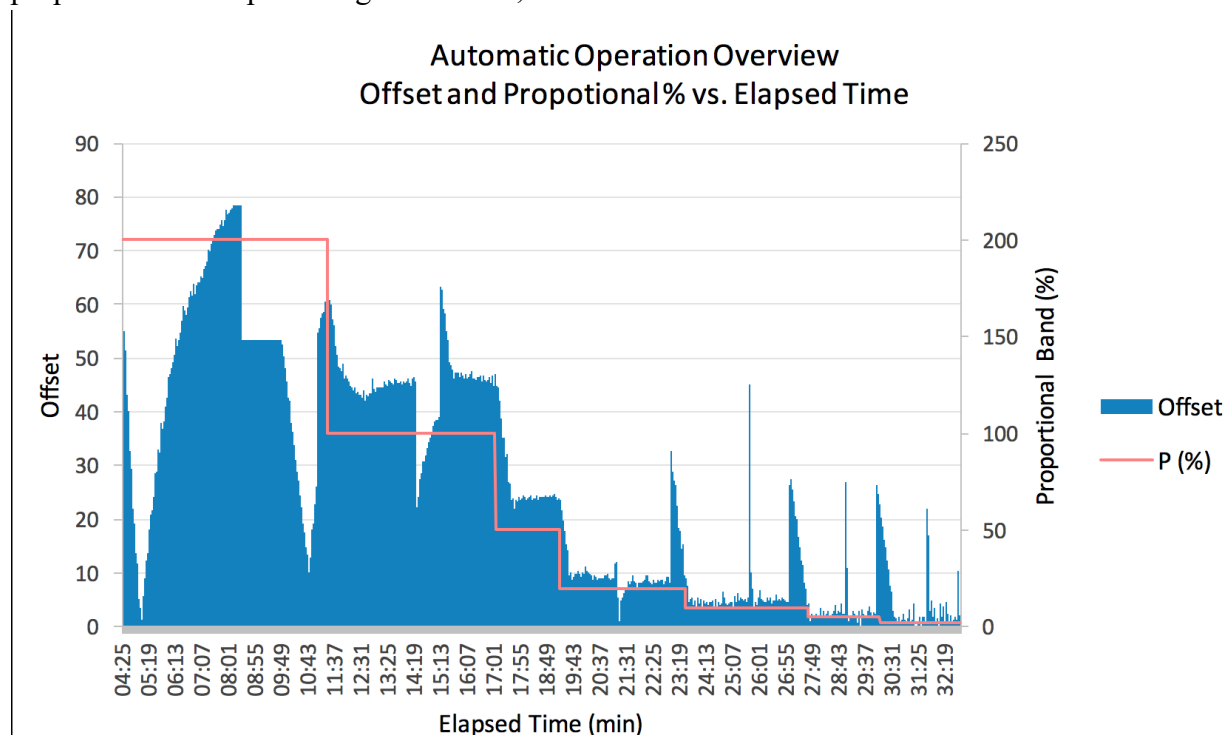
Graph 3 summarizes all of the proportional percentages as time elapsed for the P configuration. It is possible to notice how it goes from a stable flow (at 200%) to more turbulent and unstable (at 2%) as the proportional band % decreases. As seen in graph 3, whenever the solenoid valve is opened, the water level decreases and so does the set point; whereas as the solenoid valve is closed, the water level rises and the set point does too.

These data can be summarized in the following table, in which the set point offset, the proportional gain (K_p), the proportional band %, and the stability of the water level are compared.

Table 1 – Water Level Stability Compared to the Proportional Gain and the Set Point Offsets

Proportional Band Percentage (%)	Proportional gain (Kp)	Set Point Offset	Water Level Stability
2	50	5	Very Unstable
5	20	5	Very Unstable
10	10	6	Unstable
20	5	12	Semi-Stable
50	2	27	Stable
100	1	46	Very Stable
200	0.5	49	Very Stable

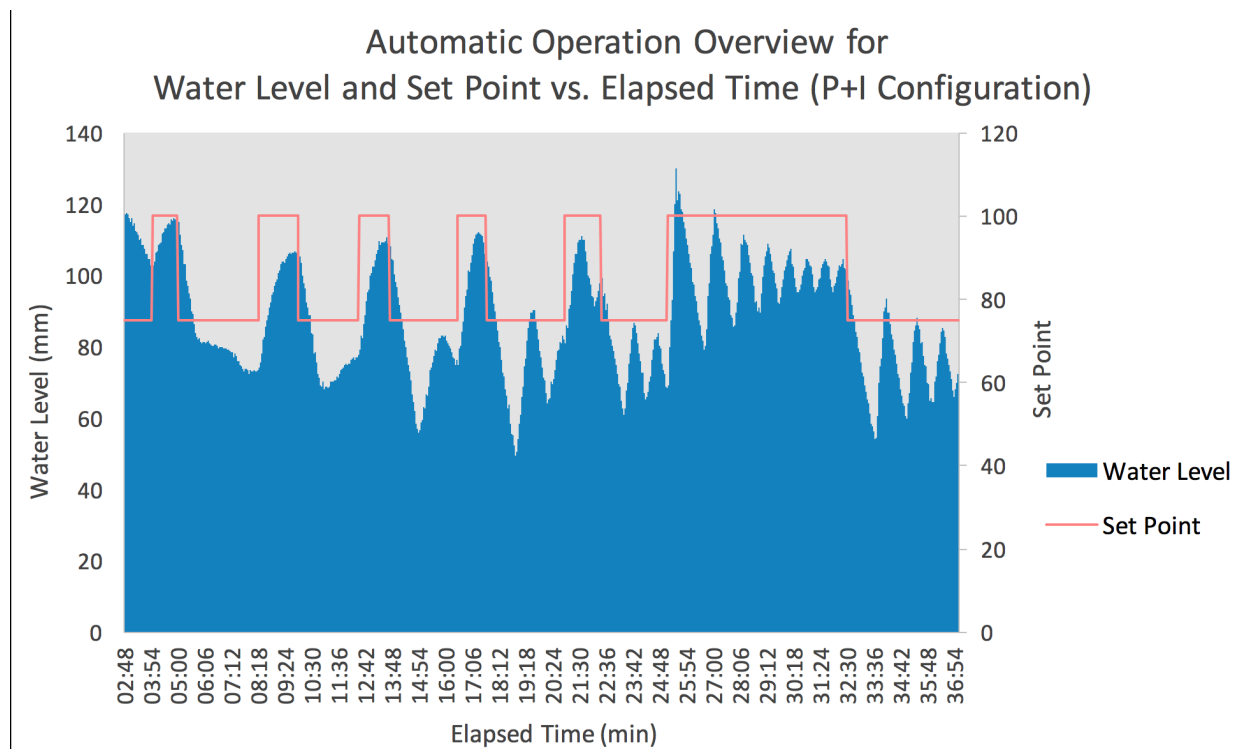
The results on this table can also be conveyed in graph 4, which displays how as the proportional band percentage decreases, the offset decreases as well with a less stable flow.

**Graph 4 - Offset and Proportional Band Percentages vs. Elapsed Time**

As seen in the graph, there are some dips within the different offsets of the process. These dips represent when the valve was open since the water flow decreased. Similarly, as the

solenoid valve was closed, the offset peaked back up but not with the same efficiency as it would along with the largest proportional band percentage.

For the P+I configuration, there were six different integral time duration that had to be carried out at a constant proportional band of 100% ($K_p = 1$). These integral time duration were measured at 2s, 5s, 10s, 20s, 50s, and 100s. For these integral duration, the set point was set up to be at 75 mm for the level sensor to reach steady state. After the steady state was reached out, the set point was changed to be at 100 mm. Finally, when the level of water reached the next level, the set point was decreased to 75 mm again for the integral time duration.



Graph 5 - Water Level and Set Point vs. Elapsed Time for the P+I Configuration

As it can be seen in graph 5, the water level oscillates more towards the end and it also has dips between the set points. The main reason why it has more oscillations towards the end is because when the integral time setting is increased, the speed of the offset gets reduced and the system becomes unstable for quick response to change the controller. Therefore, oscillation increases as time progresses because the integral time is also increased as time goes by.

5. Discussion

Exercise A:

Manually adjusting the water level by manipulating the pump speed was tedious and was prone to human error when compared to the automatic adjustment of the pump. The data that was collected was consistent with the expected results according to the theory behind the lab. When opening and closing the solenoid valve we noticed how the disturbance changed at varying pump speeds. When the pump speed was set to 33% the water level took longer to stabilize as compared to when the pump speed was set to 50% and higher. The higher the pump speed the faster the stabilization of the water level. Also, draining increased when opening the solenoid valve as compared to when the solenoid valve was closed.

Exercise C:

After analyzing the results, we can conclude that the proportional gain portion of the lab manual followed the expected trend for the control system response. The higher the proportional band the slower the system control response. Also, we noticed that at low proportional band the system became unstable, also relating to the theory in the lab manual. At high proportional bands we can see that this results in slow system response and offset. From equation 1, we can then conclude that a higher proportional gain will yield a lower K_p value.

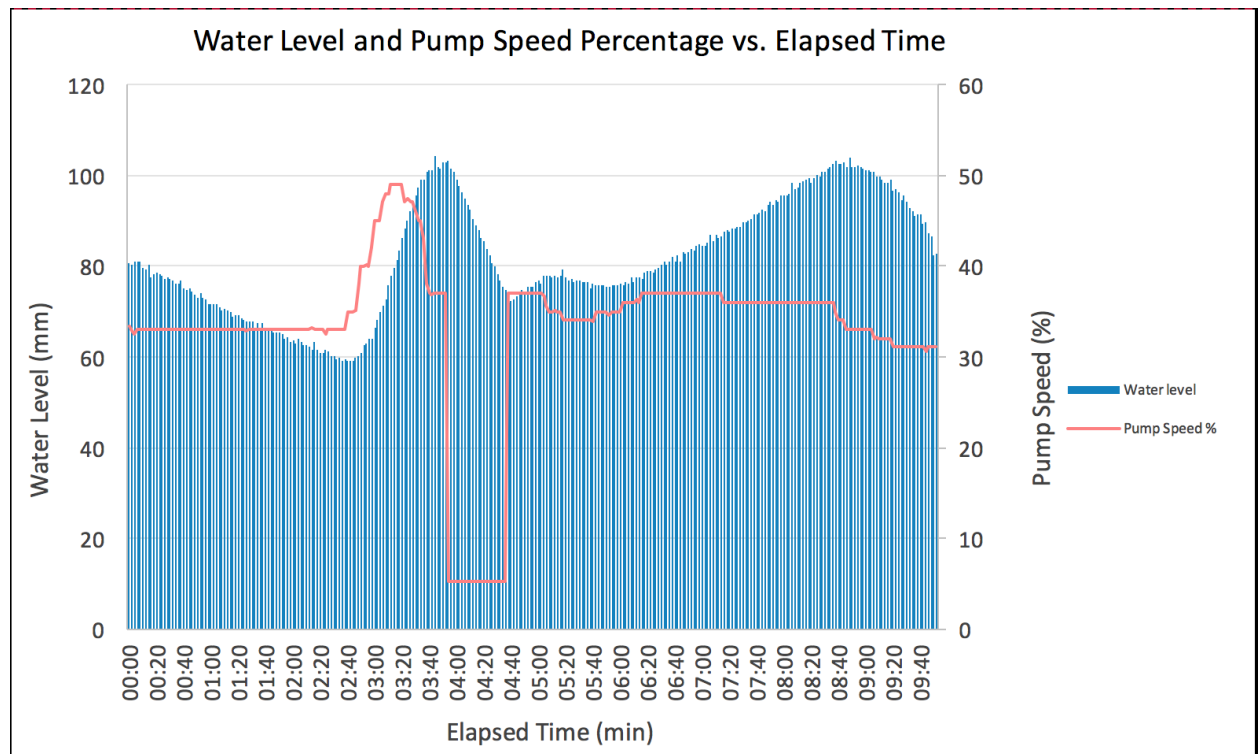
For the integral term, we tried to optimize it by making sure it is neither too long or too short. High integral term means low offset; however it means a high amount of time. On the other hand, low integral term value means high offset but the time it takes to reach set level is lowered dramatically. Therefore, it was important to optimize the value of the integral term in this experiment to make sure the set level was reached at a reasonable amount of time and at the same time to make sure the offset was not too high.

6. References

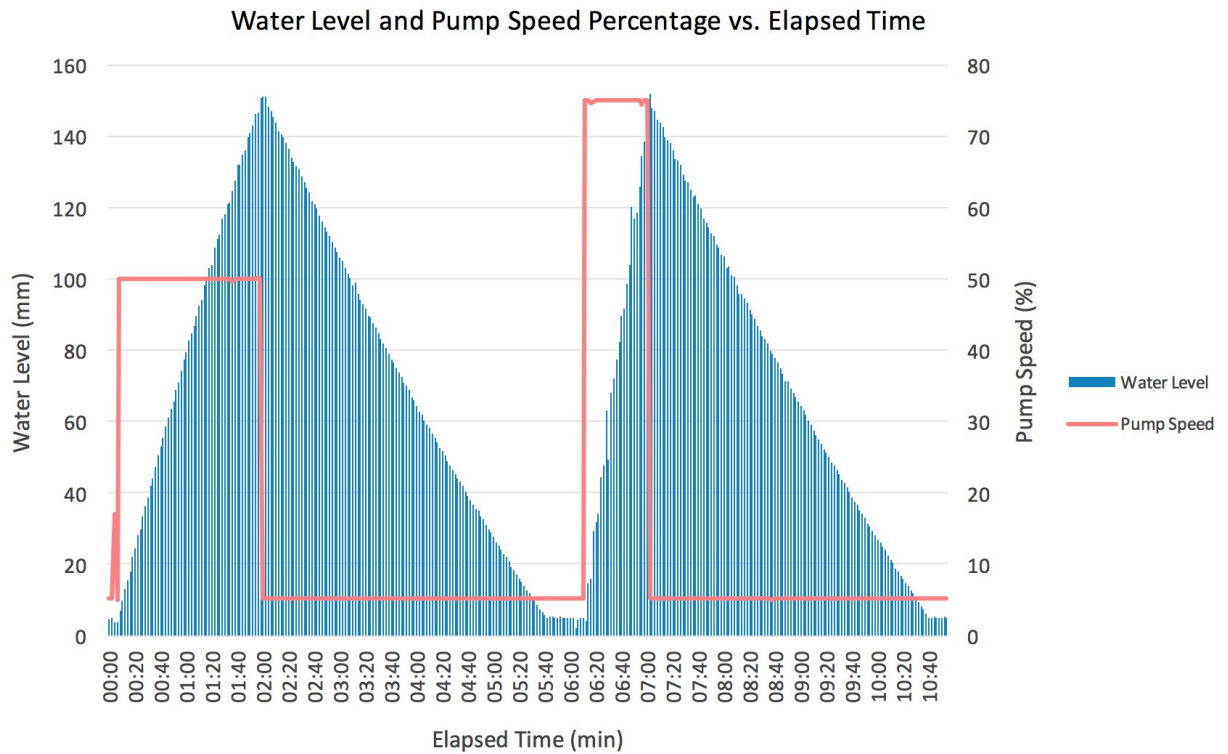
1. PROCESS CONTROL: LEVEL CONTROL Lab Manual, Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL.
2. Armfield. (n.d.). Retrieved from <http://discoverarmfield.com/en/products/view/pct50/essentials-of-process-control>.
3. US4903192A - Pid controller system. (n.d.). Retrieved from <https://patents.google.com/patent/US4903192A/en>.
4. Krishnaswamy, K. (2011). *Process control*. Tunbridge Wells: Anshan Ltd.

5. Seborg, D. E., Edgar, T. F., Mellichamp, D. A., & Doyle, F. J. (2017). Process dynamics and control. Hoboken, NJ: Wiley.
7. **Appendix I: Data Tabulation/Graphs**

Graphs for Procedure A:



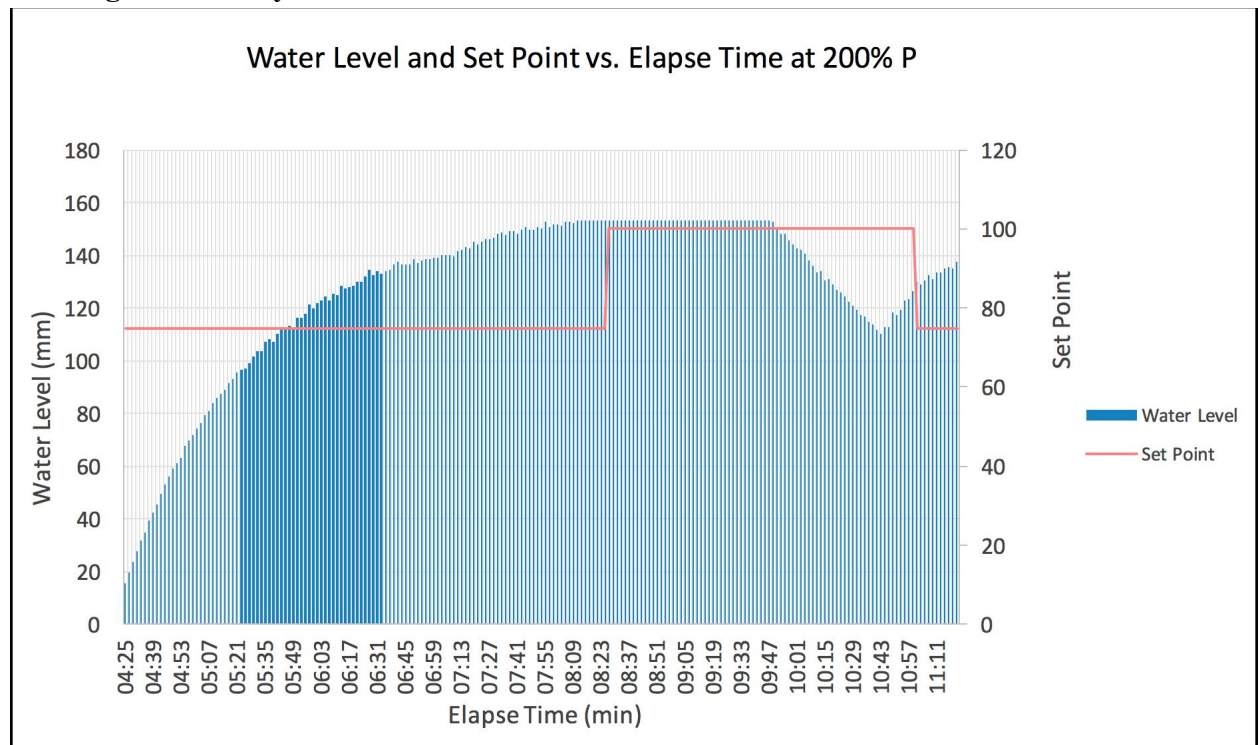
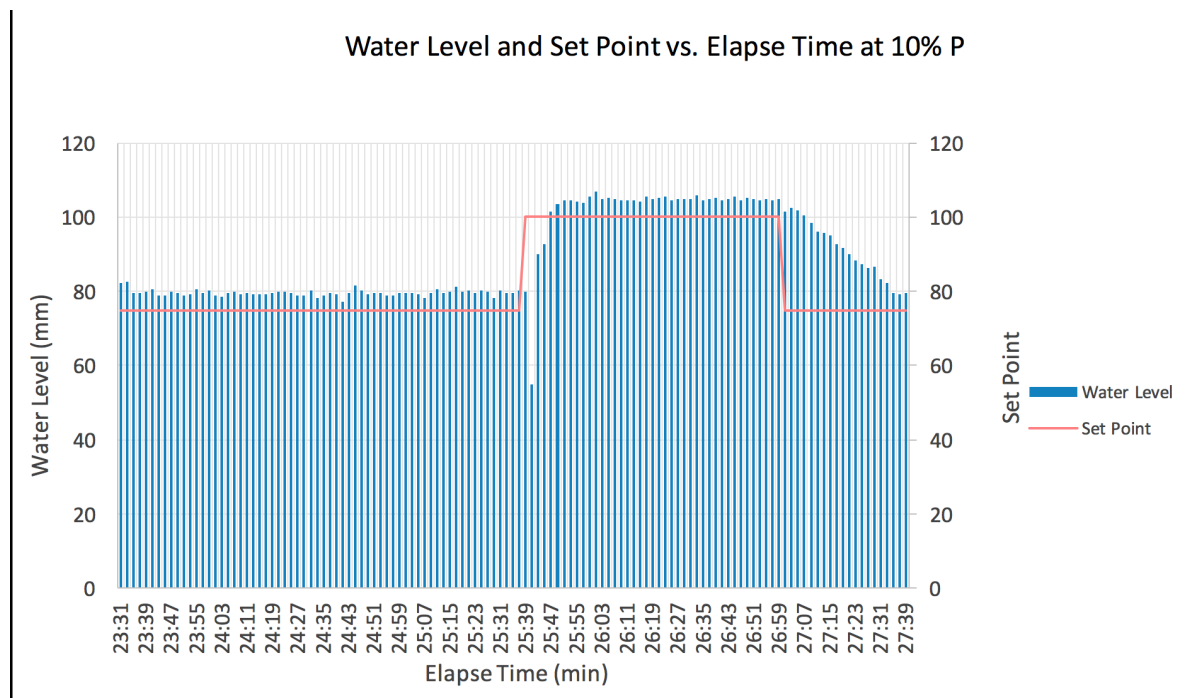
Graph 6 - Water Level as a Function of Time Elapsed for Manually Controlled Pump at Initial Fill of 80 mm

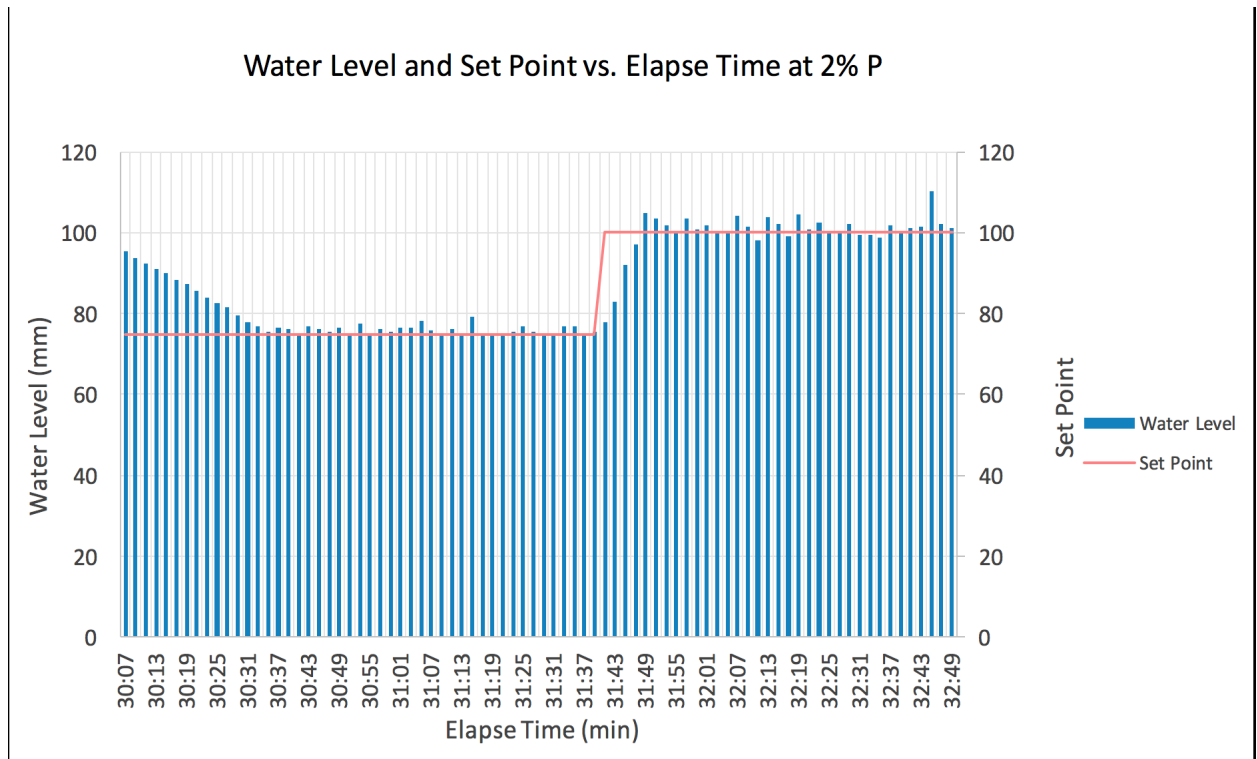


Graph 7 - Water Level as a Function of Time Elapsed for Manually Controlled Pump at Initial Fill of 0 mm

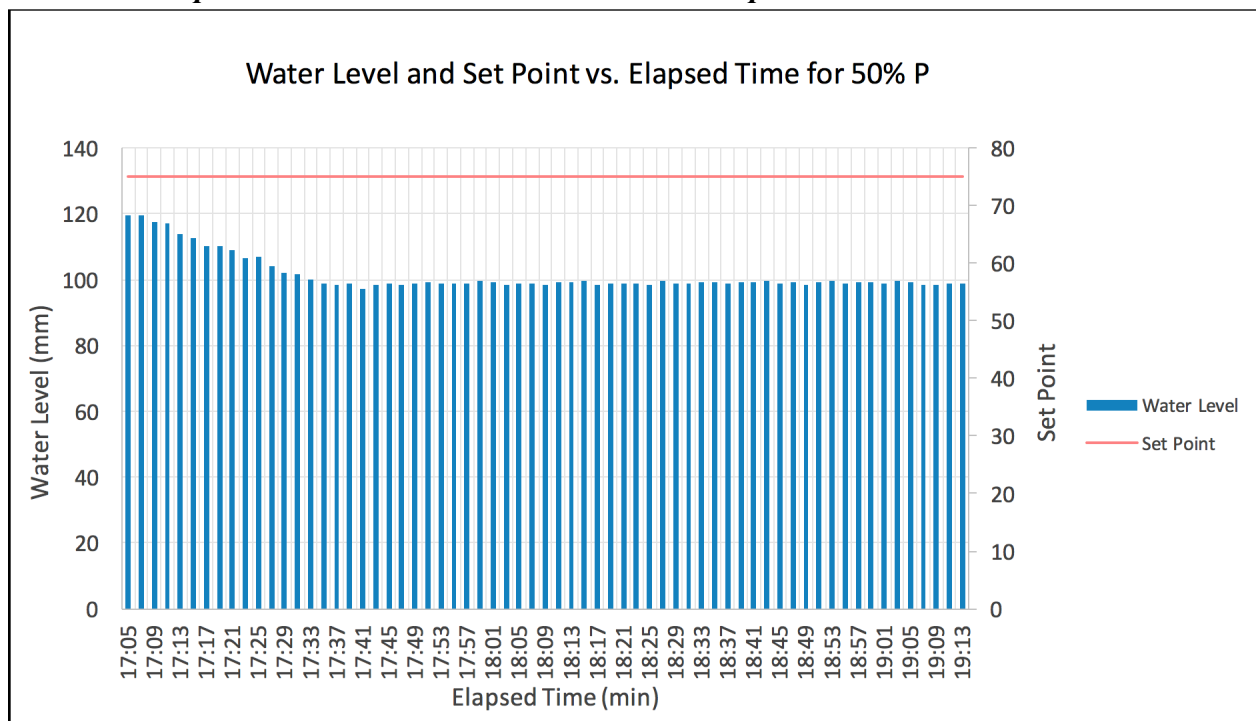
Graphs for Procedure C:

P configuration only:

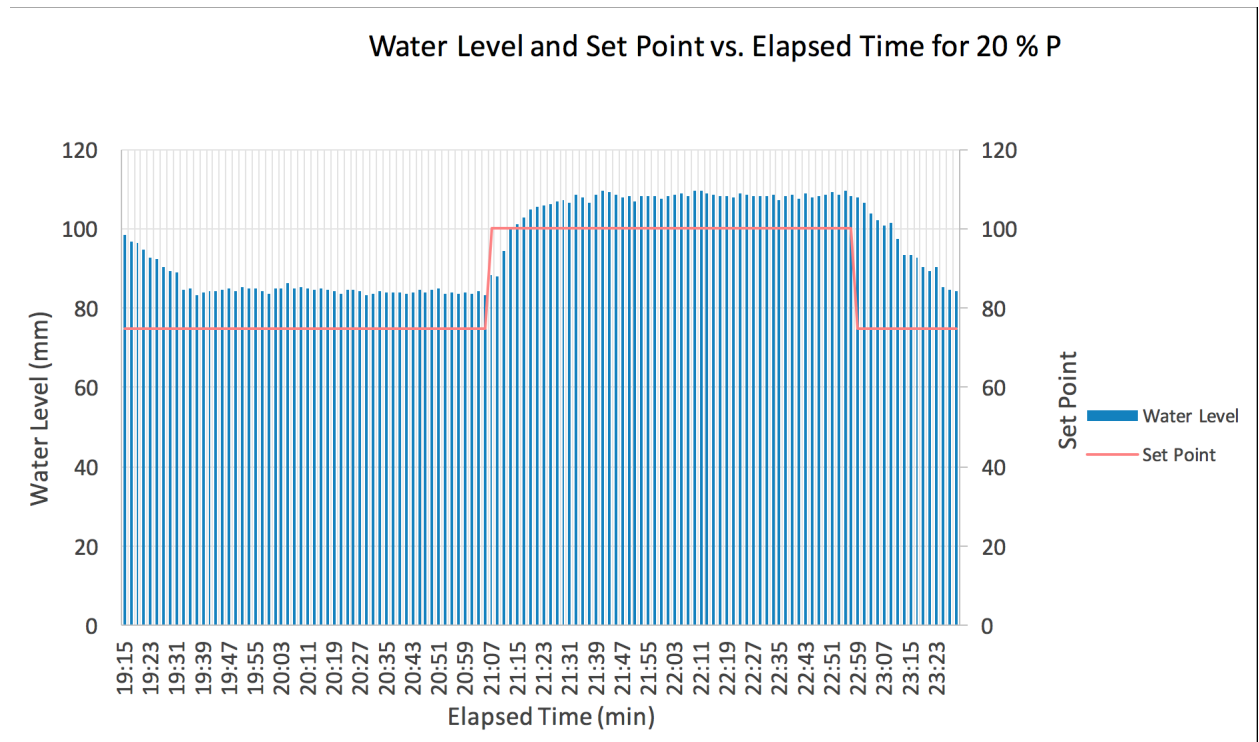
**Graph 8 - Water Level and Set Points vs. Elapsed Time at $P = 200\%$** **Graph 9 - Water Level and Set Points vs. Elapsed Time at $P = 10\%$**



Graph 10 - Water Level and Set Points vs. Elapsed Time at $P = 2\%$

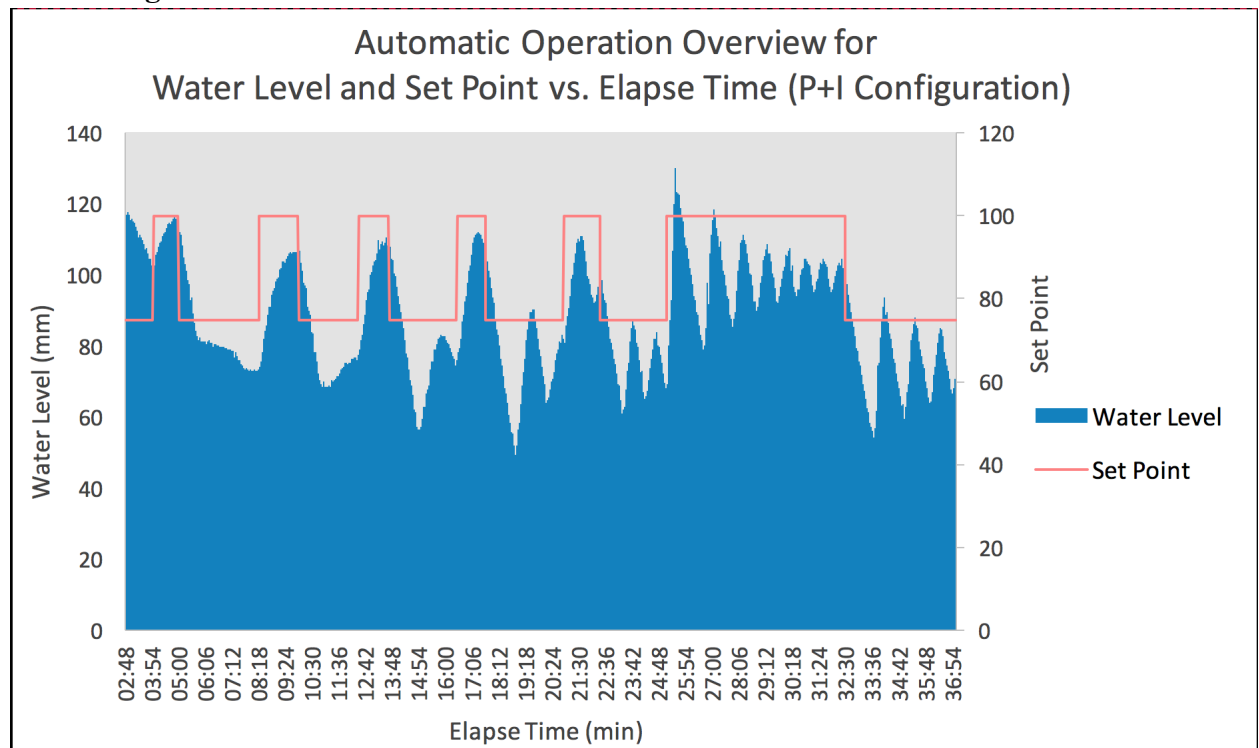


Graph 11 - Water Level and Set Points vs. Elapsed Time at $P = 50\%$

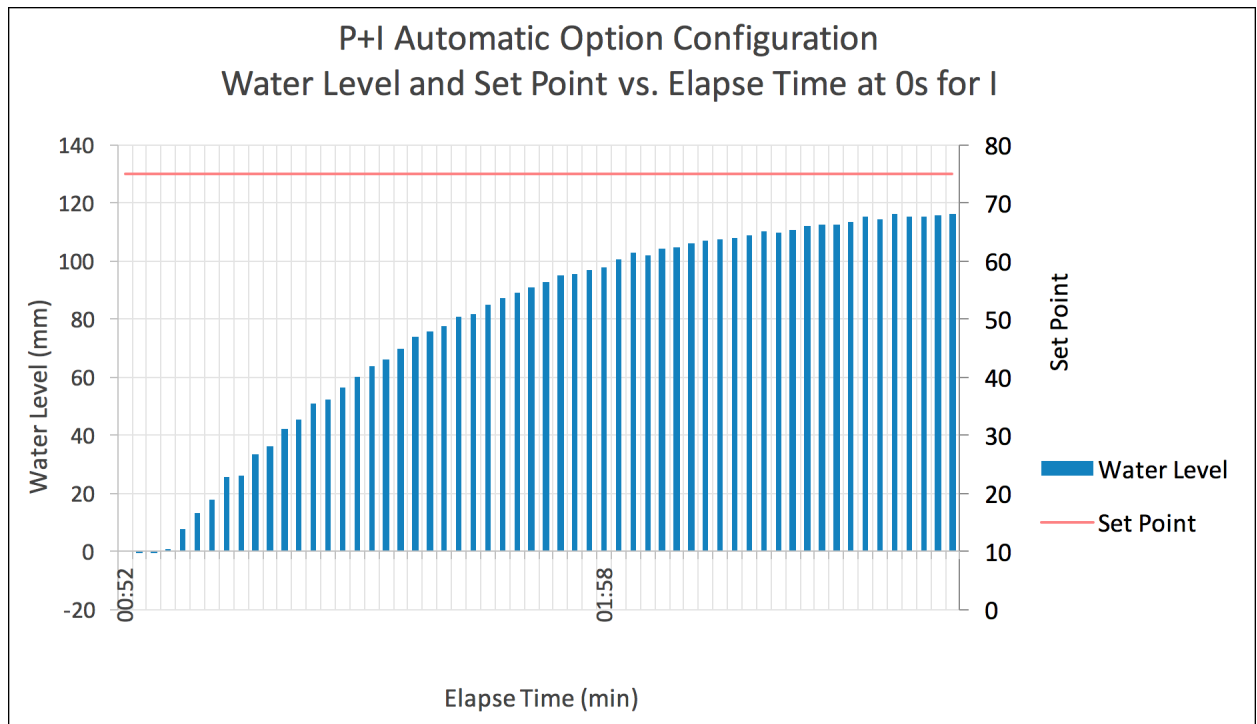


Graph 12 - Water Level and Set Points vs. Elapsed Time at $P = 20\%$

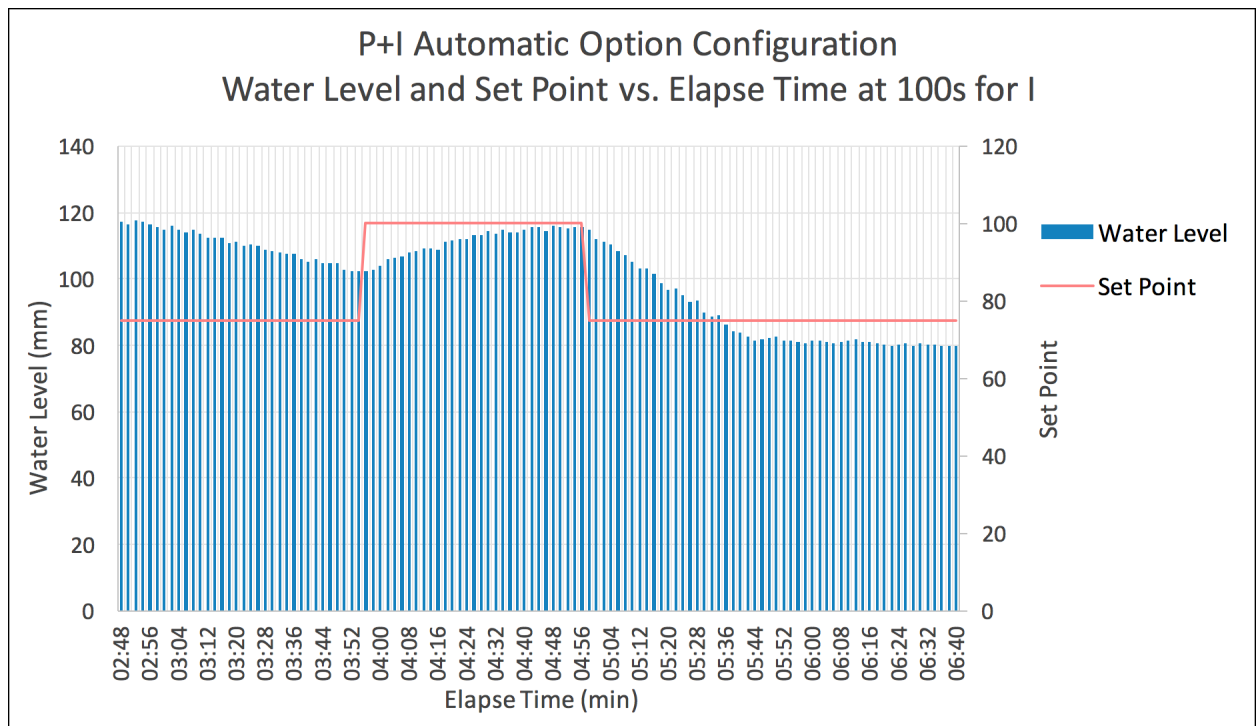
P+I Configuration:



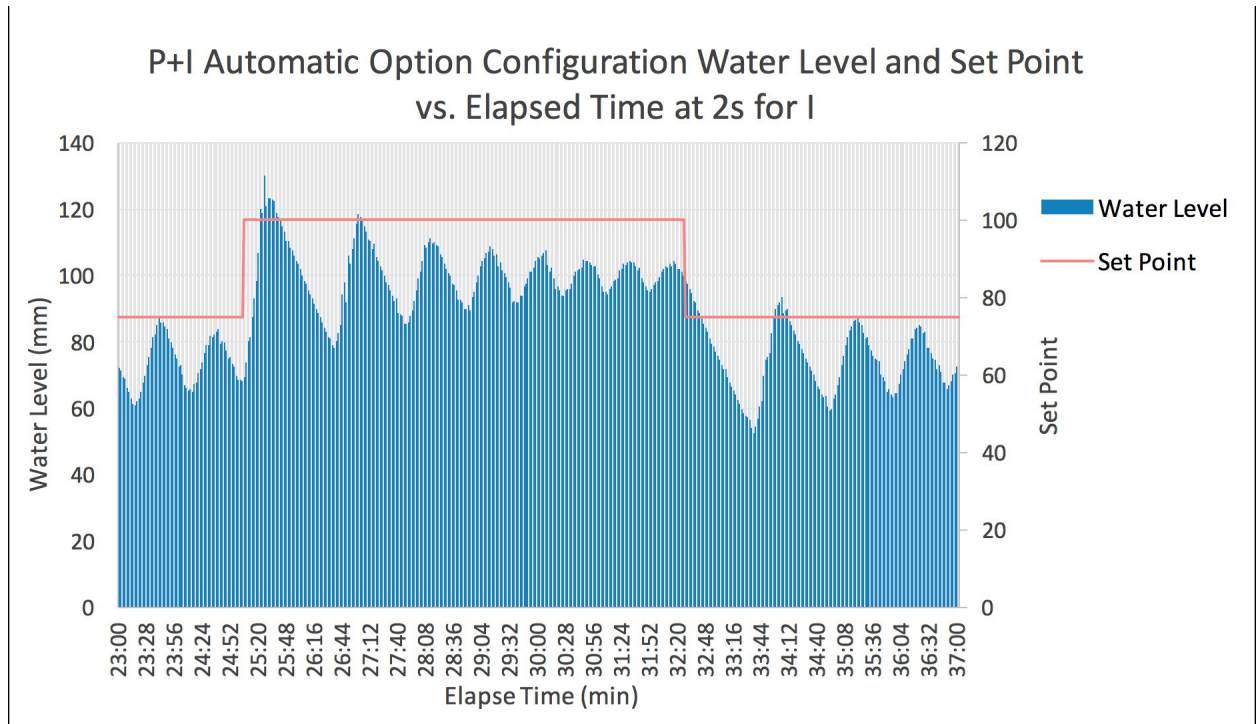
Graph 13 - Automatic Operation for the P+I Configuration Set Point at all Times



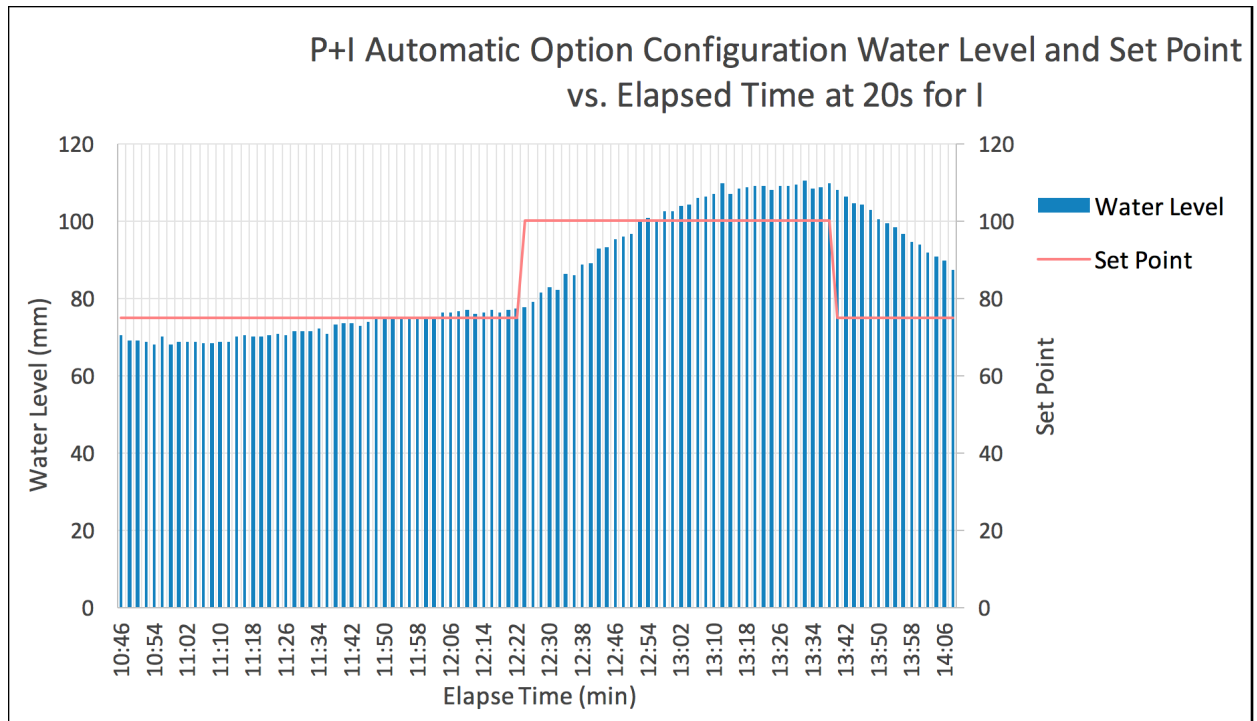
Graph 14 - Automatic Operation for the P+I Configuration Set Point at $I = 0s$



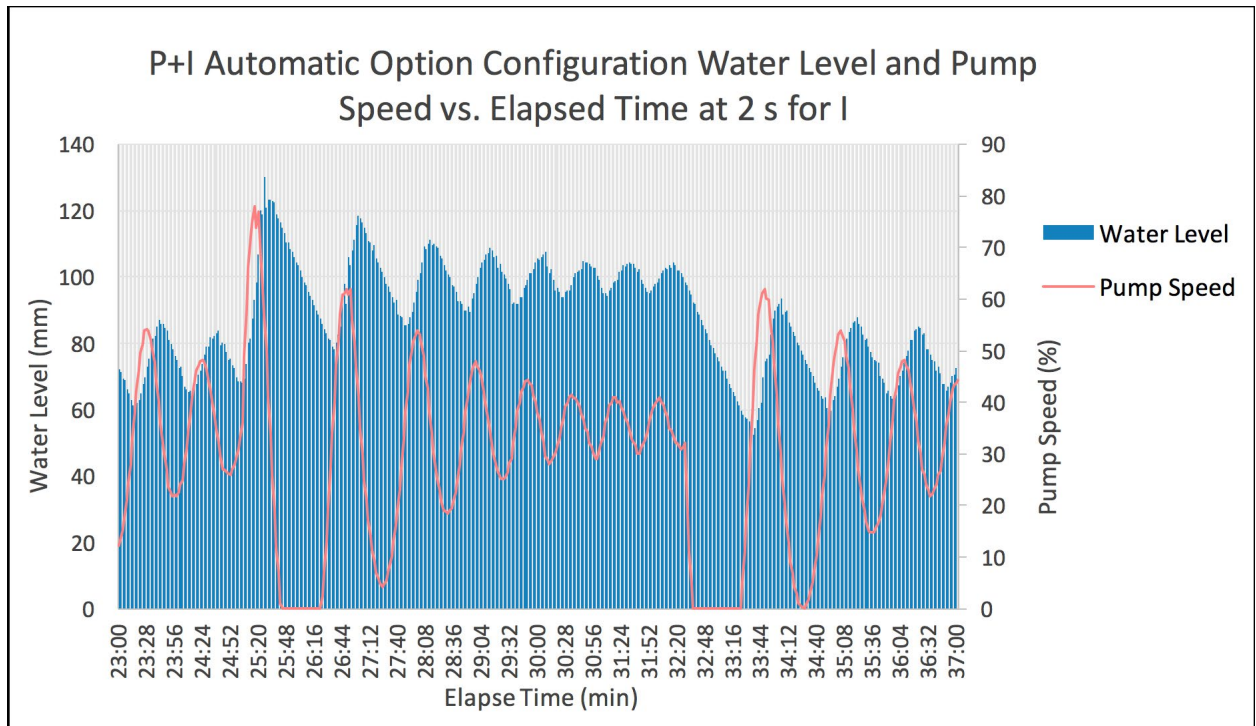
Graph 15 - Automatic Operation for the P+I Configuration Set Point at $I = 100s$



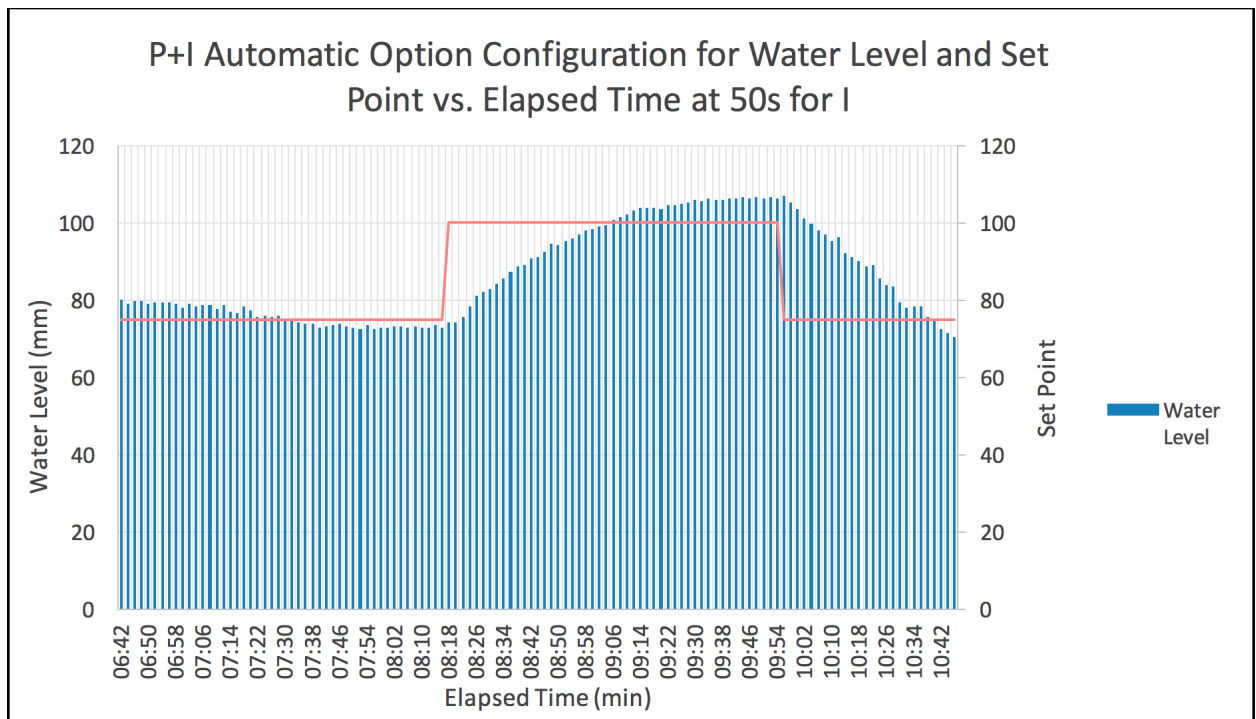
Graph 16 - Automatic Operation for the P+I Configuration Set Point at $I = 2s$



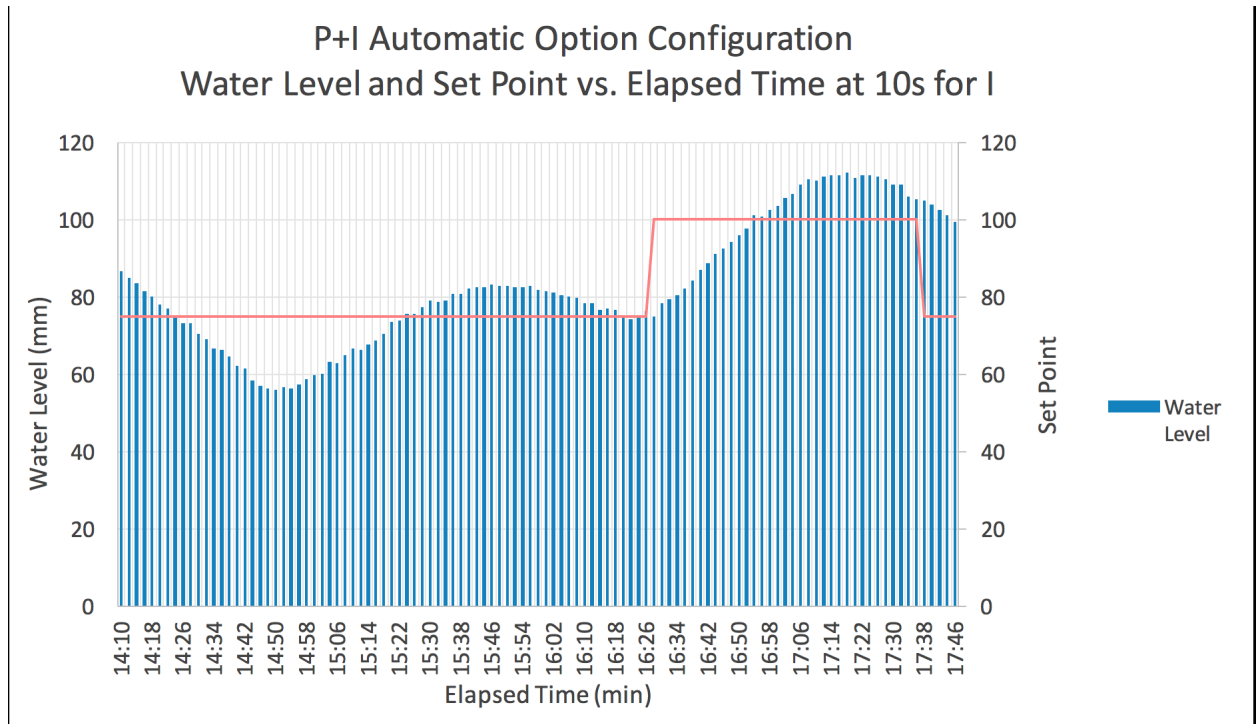
Graph 17 - Automatic Operation for the P+I Configuration Set Point at $I = 20s$



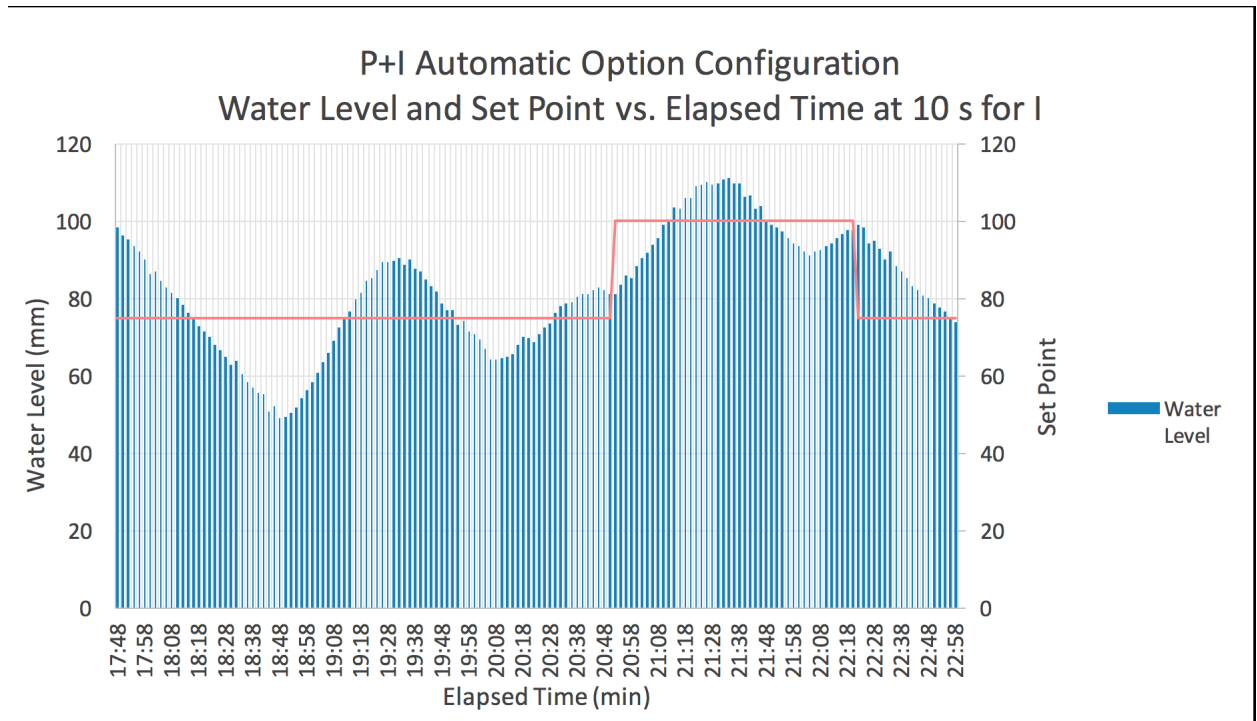
Graph 18 - Automatic Operation for the P+I Configuration Pump Speed at $I = 2$ s



Graph 19 - Automatic Operation for the P+I Configuration Set Point at $I = 50$ s



Graph 20 - Automatic Operation for the P+I Configuration Set Point at $I = 10$ s



Graph 21 - Automatic Operation for the P+I Configuration Set Point at $I = 10$ s

8. Appendix II: Error Analysis

Initially, during experiment A the speed of the pump was increased to reach 80mm. Once at the desired level, we manually matched the outflow by restricting the valve discharging from the pump. We were successful with maintaining the level in the process tank at 80mm; however, the error arose later in the laboratory. Following the directions provided in the laboratory we did not disengage the restricting valve. This visibly significantly reduced the strength of the pump system and perhaps hindered the water level from reaching the desired level in adequate time. This was especially noticeable when the pump speed was increased to 75%.

For the second portion of the experiment in exercise C, we noticed many different possible errors. The time required to change the values inputted to the PID possibly resulted in an error with the process tank level. During experiment C, we initially struggled with the performance of the PCT50, questioning the reliability of the equipment. Following multiple attempts to maintain the process level, we developed a working knowledge of the PID interface. Overall the experiment may have a significant amount of minimal errors; however, we found this to be relatively insignificant due to the large volume of data collected during the investigation.

9. Appendix III: Sample Calculations

$$\text{Offset} = |\text{Set Point} - \text{Measured Water Level}| = \text{Average Offset}$$

$$\text{Average Offset} = \frac{\sum \text{Offset}_i}{n}$$

n= number of offsets

i= offset at time t

Proportional Gain (K_p):

$$K_p = \frac{100\%}{\text{Proportional Band \%}}$$

$$K_p = \frac{100\%}{100\%}$$

$$K_p = 1$$

10. Individual Team Contributions

Name: Javier Eduardo Mendieta	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Record data/ Measured
Pre-Lab Editing	2	Contributed
Experimental Objective (Prelab)	1	Contributed
Apparatus (prelab)	1	Contributed
Materials & Supplies (prelab)	1	Contributed
Procedure (prelab)	2	Contributed
Project Safety Evaluation (prelab)	2	Contributed
Final Lab Editing	2	Contributed
Executive Summary (final lab)	2	Contributed
Introduction (Final lab)	3	Contributed
Theory (Final Lab)	3	Contributed
Results (final lab)	2	Contributed
Discussion (final lab)	2	Contributed
Conclusion (final lab)	2	Contributed
References (final and/or prelab)	1	Contributed

Data Tabulation/Graphs (final)	2	Completed
Error Analysis (final lab)	1	Contributed
Sample Calculations (final Lab)	3	Completed
PowerPoint Presentation	1	Contributed
Total	37	

Name: Omar Zahdan	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	Record data/ Measured
Pre-Lab Editing	2	Contributed
Experimental Objective (Prelab)	2	Contributed
Apparatus (prelab)	2	Contributed
Materials & Supplies (prelab)	2	Contributed
Procedure (prelab)	2	Completed
Project Safety Evaluation (prelab)	1	Contributed
Final Lab Editing	3	Contributed
Executive Summary (final lab)	2	Contributed
Introduction (Final lab)	2	Contributed

Theory (Final Lab)	1	Contributed
Results (final lab)	2	Contributed
Discussion (final lab)	2	Contributed
Conclusion (final lab)	2	Contributed
References (final and/or prelab)	1	Contributed
Data Tabulation/Graphs (final)	1	Contributed
Error Analysis (final lab)	2	Completed
Sample Calculations (final Lab)	2	Contributed
PowerPoint Presentation	1	Contributed
Total	34	

Name: Omar Takrouri	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	Record data/ Measured
Pre-Lab Editing	3	Contributed
Experimental Objective (Prelab)	1	Contributed
Apparatus (prelab)	2	Contributed
Materials & Supplies (prelab)	1	Contributed

Procedure (prelab)	1	Contributed
Project Safety Evaluation (prelab)	1	Contributed
Final Lab Editing	1	Contributed
Executive Summary (final lab)	1	Contributed
Introduction (Final lab)	2	Completed
Theory (Final Lab)	1	Contributed
Results (final lab)	2	Contributed
Discussion (final lab)	2	Completed
Conclusion (final lab)	2	Contributed
References (final and/or prelab)	3	Contributed
Data Tabulation/Graphs (final)	2	Contributed
Error Analysis (final lab)	1	Contributed
Sample Calculations (final Lab)	1	Contributed
PowerPoint Presentation	2	Contributed
Total	31	

Name: Joel Finlon	TIME (HOURS)	DESCRIPTION
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Operator (Both Lab Days)	2	Record data/ Measured
Pre-Lab Editing	1	Contributed
Experimental Objective (Prelab)	1	Contributed
Apparatus (prelab)	1	Contributed
Materials & Supplies (prelab)	1	Contributed
Procedure (prelab)	2	Contributed
Project Safety Evaluation (prelab)	1	Contributed
Final Lab Editing	4	Contributed
Executive Summary (final lab)	2	Contributed
Introduction (Final lab)	2	Contributed
Theory (Final Lab)	2	Completed
Results (final lab)	1	Contributed
Discussion (final lab)	1	Contributed
Conclusion (final lab)	2	Completed
References (final and/or prelab)	2	Contributed
Data Tabulation/Graphs (final)	1	Contributed
Error Analysis (final lab)	2	Contributed

Sample Calculations (final Lab)	1	Contributed
PowerPoint Presentation	1	Contributed
Total	30	